ANUBHAV SEN

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EDUCATION

Bachelor of Science in Computer Engineering

The Pennsylvania State University, University Park, PA — Graduation: May 2026

College of Engineering | Schreyer Honors College — Cumulative GPA: **3.94/4.0**

TECHNICAL SKILLS

- Proficient: C, Java, Python, Dash, Git, Android SDK, Android Studio, Google Firebase, XML, Verilog
- Intermediate: SQL, JavaScript, Node.js, MongoDB, React Native, Shiny

WORK EXPERIENCE

Cybersecurity Intern (SEPTA) — **June 2026 – August 2026**

- Identified **2,000+ instances of employee data** exposed across public sources by running open-source intelligence (OSINT) scans with theHarvester and SpiderFoot, mapping the org's external attack surface to reduce the risk of future phishing attacks.
- Applied spaCy for named entity recognition and Levenshtein distance matching to automatically process scraped employee data, identifying and merging roughly **30% of scraped entries** as duplicates from overlapping OSINT sources.
- Grouped scraped records into **18 clusters** using TF-IDF vectorization and K-means clustering, organizing findings by department and job function to help determine which vulnerable teams had the highest concentration of exposed data.

Global Data Operations Intern (Merck Sharp & Dohme) — **June 2024 – August 2024**

- Developed a streamlined, user-friendly dashboard to centralize data automation tools, significantly improving accessibility and usability while **reducing the time spent running manual tasks by 30%** for **over 500 users globally**.
- Created a one-click interface for running these tools, garnering strong support from non-technical users and eliminating the need for any prior experience using the command-line shell or knowing system-level commands.
- Incorporated Dash framework and collaborated with IT to secure access for deploying the site on AWS server.

SOFTWARE DEVELOPMENT PROJECTS

React Native Development — **May 2023 – Present**

- ***Nittany Retails***: e-commerce (JavaScript-based) mobile application — **August 2023**
  - Architected and hosted a Node.js/Express backend with MongoDB for schema-less storage of listings, users, and orders.
  - Integrated the Stripe API to enable secure payment processing, generating internal revenue through in-app transactions.

Android Application Development — **August 2019 – Present**

- Published **3 Android (Java-based) mobile applications** to Google Play Store that have a growing user base while currently exploring new ways to boost engagement and iteratively improve features based on user feedback.
- ***Smart Planner***: digital student planner application — **July 2022**
  - Implemented a student planning tool with automated reminders and calendar syncing (via the Google Calendar API), giving students a way to easily manage course assignments and deadlines across devices.
- ***TalkZone***: video-calling/casting media application — **October 2021**
  - Engineered real-time video calling and cross-device media casting, combining an open-source video conferencing SDK with the native Google Cast API to extend playback from mobile devices to TV screens.
- ***TagOut!***: social media application — **July 2021**
  - Developed key features such as phone-based authentication, live messaging, and personalized user feeds, powered by Google Firebase's real-time database and cloud functions for reliable performance at scale.

RESEARCH EXPERIENCE

Improving circRNA Detection: A Study of Paired-End Read Patterns — **February 2025 – Present**

- Implementing a circular RNA (circRNA) assembly framework that jointly leverages chimeric and outward reads to more accurately identify back-splicing junctions (a hallmark feature of circRNAs), overcoming blind spots in tools that rely on either read type alone.

AI for Politics — **November 2022 – May 2024**

- Predicted future US election outcomes by scraping 10K tweets/day (X API) and training a logistic-regression text classifier using TF-IDF features to label political leaning, then aggregated results into a color-coded heat map in Folium to showcase regional trends.
- Assessed model predictions against **3 past elections**, achieving directional accuracy $\leq$ **5 percentage points** of actual outcomes.

LEADERSHIP/ACTIVITES

AlgoPSU Director — **January 2023 – May 2026**

- Mentored **25 students** on system design principles and walked through LeetCode-style questions to build interview readiness.

Java: A Comprehensive Guide (Parts I & II) — **October 2021 – August 2022**

- Authored and published a two-part written guide giving a primer on fundamental data structures, algorithms, and core programming concepts in Java; only available on Amazon as a Kindle eBook, hardcover, and paperback.